COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S3
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ACTIVITY NO. 1	**Review of C++ Programming**

1. PROCEDURE OUTPUT

ILO A

1. Create a function that will take two numbers and display the sum.

```cpp
//Create a function that will take two numbers and display the sum.
#include <iostream>

//declare function SumofTwoNum as int
int SumOfTwoNum(int SumA, int SumB);


//main function to add two numbers assigned to each variable
int main() {
    std::cout <<"The sum of the two numbers is: " <<SumOfTwoNum(6,7);
}

//function to add two numbers and return the total sum
int SumOfTwoNum(int SumA, int SumB) {
    int TotalSum = SumA + SumB;
    return TotalSum;
}
```

Output:

```
The sum of the two numbers is: 13
--------------------------------
Process exited after 0.1419 seconds with return value 0
Press any key to continue . . .
```

2. Create a function that will return whether variable A is greater than variable B.

```cpp
//Create a function that will return whether variable A is greater than variable B.

//import algorithm module to use max() function
#include <iostream>
#include <algorithm>

//declare function VarAGreaterThanVarB as int
int VarAGreaterThanVarB (int VarA, int VarB);

int main() {
    //assign paramters with values
    int VarA = 21;
    int VarB = 69;

    //if-else statement that analyzes the greater variable and determine whether it's VarA or not
    VarAGreaterThanVarB (VarA, VarB);
    if (VarAGreaterThanVarB (VarA, VarB) == VarA) {
        std::cout <<"Variable A is greater than Variable B";
    }
    else {
        std::cout <<"Variable A is NOT greater than Variable B";
    }
}

//function to determine if VarA is greater than VarB
int VarAGreaterThanVarB (int VarA, int VarB) {
    int GreaterVar = std::max(VarA, VarB);
    return GreaterVar;
}
```

Output:

```
Variable A is NOT greater than Variable B
--------------------------------

Process exited after 0.2887 seconds with return value 0
Press any key to continue . . .
```

3. Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

```cpp
//Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.
#include <iostream>

//declare function BooleanOperations as bool
bool BooleanOperations (bool BoolA, bool BoolB);

//main function to after function is done
int main() {
    std::cout <<"-----All Logical Operations-----" <<std::endl;
    BooleanOperations (true, false);
}

//function to display all logical operation as 0 or 1 and return true if successful
bool BooleanOperations (bool BoolA, bool BoolB) {
    std::cout <<"BoolA: " <<BoolA <<std::endl;
    std::cout <<"BoolB: " <<BoolB <<std::endl;
    std::cout <<std::endl;
    std::cout <<"Negation: " <<std::endl;
    std::cout <<"BoolA: " <<!BoolA <<std::endl;
    std::cout <<"BoolB: " <<!BoolB <<std::endl;
    std::cout <<std::endl;
    std::cout <<"Conjunction: " <<(BoolA && BoolB) <<std::endl;
    std::cout <<std::endl;
    std::cout <<"Disjunction: " <<(BoolA || BoolB) <<std::endl;
    return true;
}
```

Output:

```
-----All Logical Operations-----
BoolA: 1
BoolB: 0

Negation:
BoolA: 0
BoolB: 1

Conjunction: 0

Disjunction: 1

--------------------------------
Process exited after 0.1456 seconds with return value 0
Press any key to continue . . .
```

ILO C

1. Create a C++ program to swap the two numbers in different variables.

```cpp
//Create a C++ program to swap the two numbers in different variables.

//import algorithm module to the program to use swap() module
#include <iostream>
#include <algorithm>

int main() {
    //declare both variables
    int Var1 = 420;
    int Var2 = 2763;

    std::cout <<"Content of variables before the swap:" <<std::endl;
    std::cout <<"Var1: " <<Var1 <<std::endl;
    std::cout <<"Var2: " <<Var2 <<std::endl;
    std::cout <<"\n";

    //use swap() function to swap the two numbers
    std::swap(Var1, Var2);

    std::cout <<"Content of variables after the swap:" <<std::endl;
    std::cout <<"Var1: " <<Var1 <<std::endl;
    std::cout <<"Var2: " <<Var2 <<std::endl;
}
```

Output:

```
Content of variables before the swap:
Var1: 420
Var2: 2763

Content of variables after the swap:
Var1: 2763
Var2: 420

--------------------------------
Process exited after 0.1503 seconds with return value 0
Press any key to continue . . .
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```cpp
//Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

#include <iostream>

//declaring the function
double KelvinToFahrenheit(double Kelvin);

//main program to output temperature conversion
int main () {
    double Kelvin;
    std::cout <<"Please enter the temperature in Kelvin: ";
    std::cin >> Kelvin;
    std::cout <<"Temperature in Fahrenheit: " <<KelvinToFahrenheit(Kelvin) <<std::endl;
}

//function for converting Kelvin to Fahrenheit
double KelvinToFahrenheit(double Kelvin) {
    double Celsius = Kelvin - 273.15;
    double Fahrenheit = (1.8)*Celsius + 32;
    return Fahrenheit;
}
```

Output:

```
Please enter the temperature in Kelvin: 100
Temperature in Fahrenheit: -279.67

--------------------------------
Process exited after 9.624 seconds with return value 0
Press any key to continue . . .
```

3. Create a C++ program that has a function that will calculate the distance between two points.

```cpp
//Create a C++ program that has a function that will calculate the distance between two points.

//import math module into the program to use sqrt() and pow() function
#include <iostream>
#include <cmath>

//declaring the function
double DistanceOfTwoPoints(int PointX1, int PointX2, int PointY1, int PointY2);

//main function
int main() {
    //declare all parameters
    int PointX1;
    int PointY1;
    int PointX2;
    int PointY2;

    //porgram ask for users input
    std::cout <<"Please enter the x-axis of the first point: ";
    std::cin >> PointX1;
    std::cout <<"Please enter the y-axis of the first point: ";
    std::cin >> PointY1;
    std::cout <<"Please enter the x-axis of the second point: ";
    std::cin >> PointX2;
    std::cout <<"Please enter the y-axis of the second point: ";
    std::cin >> PointY2;
    std::cout <<"The distance between these two points is: " <<DistanceOfTwoPoints(PointX1, PointX2, PointY1, PointY2);
}

//function for calculating distance between two points
double DistanceOfTwoPoints(int PointX1, int PointX2, int PointY1, int PointY2) {
    double Distance = sqrt(pow(PointX2 - PointX1, 2.0) + pow(PointY2 - PointY1, 2.0));
    return Distance;
}

```

Output:

```
Please enter the x-axis of the first point: 4
Please enter the y-axis of the first point: 5
Please enter the x-axis of the second point: 2
Please enter the y-axis of the second point: 9
The distance between these two points is: 4.47214
--------------------------------
Process exited after 5.686 seconds with return value 0
Press any key to continue . . .
```

4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```cpp
//Modify the code given in ILO B and add the following functions:
//a. A function to compute for the area of a triangle
//b. A function to compute for the perimeter of a triangle
//c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

//import  algorithm module to use max() function
#include <iostream>
#include <algorithm>

//declare all three functions
double AreaOfATriangle (double BaseA, double TriangleHeight);
double PerimeterOfATriangle (double SideA, double BaseA, double SideB);
double DetermineTriangle (double A, double B, double C, double MaxAngle);

class Triangle {
    private:
        double totalAngle, angleA, angleB, angleC;
    public:
        Triangle(double A, double B, double C);
        void setAngles(double A, double B, double C);
        const bool validateTriangle();
};

int main(){
    //declare paramters
    double BaseA;
    double TriangleHeight;
    double SideA;
    double SideB;
    double A = 40;
    double B = 30;
    double C = 110;
    double MaxAngle;
    //driver code
    Triangle set1(40, 30, 110);
        std::cout <<"The shape is a valid triangle. \n";
    }
    else {
        std::cout <<"The shape is NOT a valid triangle. \n";
    }
    //ask user for input for base, height, and both sides
    std::cout <<"Please enter the length of the base of the Triangle: ";
    std::cin >> BaseA;
    std::cout <<"Please enter the length of the height of the Triangle: ";
    std::cin >> TriangleHeight;
    std::cout <<"Please enter the length of the side A of the Triangle: ";
    std::cin >> SideA;
    std::cout <<"Please enter the length of the side B of the Triangle: ";
    std::cin >> SideB;
    std::cout <<"The area of the triangle is: " <<AreaOfATriangle (BaseA, TriangleHeight) <<std::endl;
    std::cout <<"The perimeter of the triangle is: " <<PerimeterOfATriangle (SideA, BaseA, SideB) <<std::endl;
    //if-else statmeent to determine the type of triangle
    DetermineTriangle (A, B, C, MaxAngle);
    if (DetermineTriangle (A, B, C, MaxAngle) < 90) {
        std::cout <<"This triangle is acute-angled";
    }
    else if (DetermineTriangle (A, B, C, MaxAngle) == 90) {
        std::cout <<"This triangle is right-angled";
    }
    else if (90 < DetermineTriangle (A, B, C, MaxAngle) < 180) {
        std::cout <<"This triangle is obtuse-angled";
    }
    else {
        std::cout <<"This is not a triangle";
    }
    return 0;
}

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
      angleC = C;
      totalAngle = A+B+C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle <=180);
}
//function to find the area of a Triangle and return its area
double AreaOfATriangle (double BaseA, double TriangleHeight) {
    double area = (BaseA*TriangleHeight)/2.0;
    return area;
}
//function to find the perimeter of a Triangle and return the perimeter
double PerimeterOfATriangle (double SideA, double BaseA, double SideB) {
    double perimeter = SideA + BaseA + SideB;
    return perimeter;
}
//fucntion to determine the type of Triangle and return the greater angle
double DetermineTriangle (double A, double B, double C, double MaxAngle) {
    MaxAngle = std::max(A, B);
    MaxAngle = std::max(MaxAngle, C);
    return MaxAngle;
}
```

Output:

```
The shape is a valid triangle.
Please enter the length of the base of the Triangle: 6
Please enter the length of the height of the Triangle: 2.5
Please enter the length of the side A of the Triangle: 9
Please enter the length of the side B of the Triangle: 3
The area of the triangle is: 7.5
The perimeter of the triangle is: 18
This triangle is obtuse-angled
--------------------------------
Process exited after 20.07 seconds with return value 0
Press any key to continue . . .
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	None
Class Declaration and Method Definition Section	`class Triangle { private: double totalAngle, angleA, angleB, angleC; public: Triangle(double A, double B, double C); void setAngles(double A, double B, double C); const bool validateTriangle(); };`
Main Function	`int main(){ //driver code Triangle set1(40, 30, 110); if(set1.validateTriangle()){ std::cout <<"The shape is a valid triangle. \n"; } else { std::cout <<"The shape is NOT a valid triangle. \n"; } return 0; }`
Method Definition	`Triangle::Triangle(double A, double B, double C) { angleA = A; angleB = B; angleC = C; totalAngle = A+B+C; } void Triangle::setAngles(double A, double B, double C) { angleA = A; angleB = B; angleC = C; totalAngle = A+B+C; } const bool Triangle::validateTriangle() { return (totalAngle <=180); }`

Table 1-1. C++ Structure Code for Answer

```
The shape is a valid triangle


--------------------------------
Process exited after 0.02163 seconds with return value 0
Press any key to continue . . .
```

This output shows that the angles that was inputted into those angles shows that you can make a triangle out of those angles. The code will try to validate the following angles if the angles are equal or less than 180 degrees. The total angles will be added and determine if it's a triangle or not.

Table 1-2. ILO B output observations and comments

8. Conclusion

In conclusion, functions group lines of code into a single block. This makes the code neat and easier to understand. You can make the function either before or after the main function. In the procedure, I made a program based on what I learned from the previous C++ course. In this program, I was able to call functions, import modules such as cmath and algorithm, and use other data types such as bool. In the supplementary activity, I made both a program with and without a function, using more of the lessons I learned in my previous course. In that program, I used other data types, imported additional modules, and even called functions that involve a class. Overall, I think I did decently in the activity. There might be some mistakes and difficulties in the code, but I managed to finish all the given activities. I would like to improve at creating more programs that ask the user for input, because sometimes programs ask you to enter values, and it will be helpful for future activities.

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT and Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code Assistance and Grammar Correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used ChatGPT to assist me in debugging this code without showing the answers. I prompted that it will not give answers to ensure that I can still get my code running, without the AI doing all the work.**
I used Grammarly to fix any grammar mistakes I might have made when making the conclusion it will help to make the conclusion cleaner and easier to understand.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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